5.2 PURPOSE OF HEAT TREATMENT

1. Improvement in ductility
2. Relieving internal stresses
3. Refinement of grain size
4. Increasing hardness or tensile strength
5. Improvement in machinability
6. Alteration in magnetic properties
7. Improvement in toughness and development of re-crystallized structure.

5.3 LOWER AND HIGHER CRITICAL TEMPERATURE

While heating a solid metals, their internal structure starts to transform at a particular temperature. This temperature is known as Lower critical temperature of the metal. On further heating, the whole internal structure is transformed at particular temperature. This temperature is called upper critical temperature.



இரும்பை காய்ச்சி உருக்கிடுவீரே...
இயந்திரங்கள் வகுத்திடுவீரே...

—பாரதியார்

Melt the Iron and
Make the Machine

—Bharathiyar

5.4 METHOD OF HEAT TREATMENT PROCESS

1. Annealing
2. Normalising
3. Hardening
4. Tempering
5. Case Hardening
   a. Carburising
   b. Nitriding
   c. Cyaniding
   d. Induction Hardening
   e. Flame Hardening

ANNEALING

Annealing is a process in which a metal is heated to a particular high temperature, held there for a period of several hours or several days, and allowed to cool slowly, by using sand, lime or ashes. This process is mainly applied to produce softening.

Process of Annealing

1. In this process low carbon steel is heated to 30°C to 50°C above the higher critical temperature.
2. Maintain it in the same temperature for a considerable period of time.
3. The metal is slowly cooled by placing in to sand, ashes or lime that insures a slow rate of cooling.
4. Oil fired furnace, gas fired furnace or sintering (Electrical) furnace are used for heating.

The temperatures are monitored by thermo couple.

Purpose of Annealing

1. To soften the steel.
2. To improve machinability.
3. To increase ductility and toughness.
4. To relieve internal stresses.
5. To refine grain size.
6. To improve homogeneity.

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